12. Hill Cipher — Encryption and Decryption

CODE:

```
import numpy as np

key = np.array([[9, 4],
                [5, 7]])

message = "meet me at the usual place at ten rather than eight oclock"
message = "".join(ch for ch in message.upper() if ch.isalpha())

if len(message) % 2 != 0:
    message += "X"

# Encryption
ciphertext = ""

for i in range(0, len(message), 2):
    pair = [ord(message[i]) - 65,
            ord(message[i+1]) - 65]

    result = np.dot(key, pair) % 26

    ciphertext += chr(result[0] + 65)
    ciphertext += chr(result[1] + 65)

print("Plaintext:", message)
print("Ciphertext:", ciphertext)

# Decryption
det = int(round(np.linalg.det(key)))
det = det % 26
inverse_det = pow(det, -1, 26)

adjoint = np.array([[7, -4],
                    [-5, 9]])

inverse_key = (inverse_det * adjoint) % 26

decrypted = ""

for i in range(0, len(ciphertext), 2):
    pair = [ord(ciphertext[i]) - 65,
            ord(ciphertext[i+1]) - 65]

    result = np.dot(inverse_key, pair) % 26

    decrypted += chr(int(result[0]) + 65)
    decrypted += chr(int(result[1]) + 65)

print("Inverse key matrix:")
```

print(inverse_key)

print("Decrypted text:", decrypted)

OUTPUT:

